

ObsSec Recommendations for Summer 2026

Introduction

This document contains a selection of our recommended objects to look at with the telescopes this summer. It is by no means a comprehensive list of everything there is to see, but instead contains some of the brighter or otherwise interesting objects, and provides a good starting point for visual observing. We've included no photos of anything, since these often make your own observations pale in comparison and distract you from appreciating that you can see the objects with your *very own eyes*.

The selection of objects is organised by constellation, which are themselves organised roughly in increasing Right Ascension (RA). As a result, objects listed earlier in the guide are highest in the sky and best seen earlier in the term or earlier in the evening. Objects near the end are best seen later in the term or later at night.

For each object, there is:

- The catalog name [*Messier (M)*, *Index (IC)*, *New General (NGC)*] and any common name.
- Object class, described below.
- Our difficulty rating, explained below.
- The apparent magnitude, which corresponds to how bright the object is. A smaller magnitude means brighter, and any object brighter than ~5 can be seen with the naked eye with our levels of light pollution.
- Celestial coordinates, in RA and Dec (Declination).
- A brief description of the object, with some observing tips and specific points of interest.

Difficulty scale:

- **Easy** ▾
These objects are great to start with. They will typically be easy to find, have lots of visible features, and will jump out of the eyepiece at you.
- **Medium** ▾
These are harder to spot, requiring some effort to find and especially to make out detail. Moonless nights and turning off the dome lights will help.
- **Hard** ▾
Most of these objects need some experience to observe, including long dark adaptation and averted vision. Detail can be very tricky to discern, and often a successful observation reveals just a faint blur and the knowledge that you've seen something really exciting with your own eyes.
- **Insane or Astrophotography** ▾
These objects might well be impossible to see in Cambridge, requiring larger telescope apertures and darker skies than we have here. Do give them a go, however, and many of them are also great targets for astrophotography if you have the ability to do that.

Object Classes:

- **Planets**
Large bodies orbiting our sun. You are standing on one of them! Through a telescope, they are small and bright, so take magnification (via smaller eyepieces) well. Details on their surfaces are limited by seeing (atmospheric turbulence) instead of contrast (from light pollution).
- **Multiple star systems**
Stars that are physically close to and orbit each other. The contrast between different star types is readily apparent for closely separated ones. Some particularly tight pairs can require good seeing to 'split' (i.e. to convince yourself that the point of light is actually two stars).
- **HII regions**
Large glowing emission nebulae, primarily of hydrogen gas. These act as stellar nurseries, and the newly formed stars within them ionise the gas to glow a deep red, among other colours (which are usually only visible in photos). Often only unusually bright regions are visible visually, although filters can help increase the contrast.
- **Planetary nebulae**
Named for being small, round, relatively bright, and therefore planet-like, these emission nebulae are glowing shells of gas surrounding hot dying stars. Their small size and higher contrast make them great targets for teasing out difficult detail in a large focal length telescope like the Northumberland.
- **Supernova remnants**
Emission nebulae formed from the violent explosion of the progenitor star at the end of its life. While rare and (astronomically) short-lived, we do have the *Crab Nebula* (M1) visible with the telescopes, appearing similar to a faint planetary nebula.
- **Reflection nebulae**
Often combined with other nebula types, these are simply dust clouds reflecting broadband starlight. As a result, they appear white in photographs and are not improved with filters, sometimes necessitating darker skies than we have here.
- **Open clusters**
Groups of stars which all formed together, and younger ones are often associated with the HII region that birthed them. They appear as gatherings of tens to hundreds of easily distinguishable points.
- **Globular clusters**
Collections of thousands to hundreds of thousands of gravitationally bound stars. These are often old and distant, outside the plane of our galaxy. They can be seen as hazy globes, in which you can often discern the faint speckling of the thousands of constituent stars.
- **Galaxies**
These are the most distant objects easily seen, and are collections of tens of billions of stars and all the other objects discussed here. They are usually visible through the telescopes as elliptical clouds with a bright core, and some have subtle detail from spiral arms or dust lanes that can be picked out with practice.

Good luck observing, and clear skies!

- *The ObsSecs, 2026*

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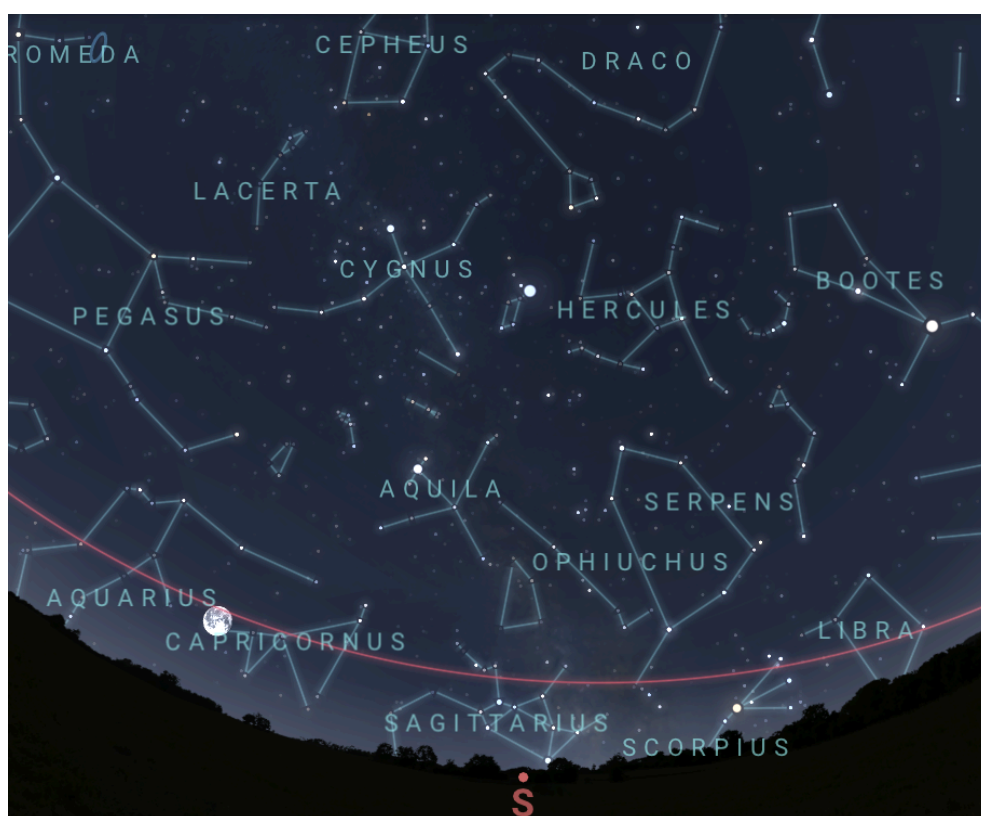
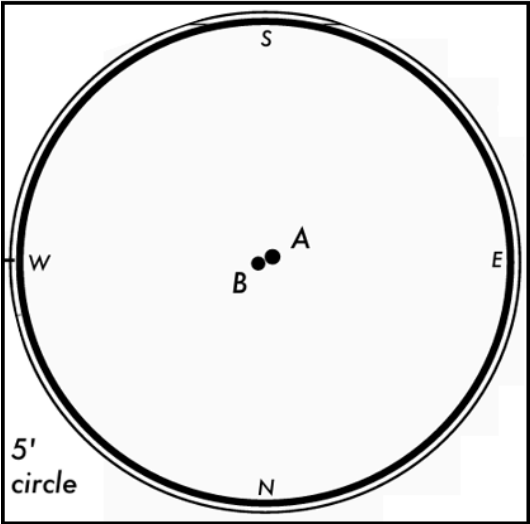


Photo credit: Stellarium. Shown is the summer sky from Cambridge, looking southwards. During the summer, the ecliptic (the red line) is positioned low in the sky at dusk, making it difficult to observe deep sky objects below it.

Note: Unless otherwise specified, figures have been taken from *'Turn Left at Orion'* - G. Consolmagno, D. M. Davis. Object descriptions were written by William Moorhouse (ObsSec 2026-27).

Corona Borealis, Serpens



Zeta CrB (Multiple Star) Easy

RA 15h 40m 22s | Dec 36° 33' 02" | Mag 4.96, 5.91

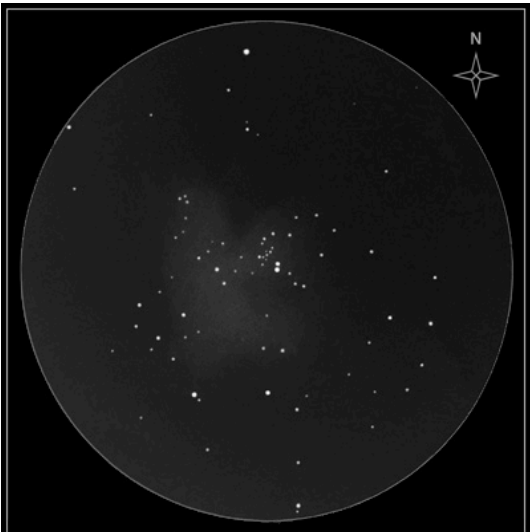
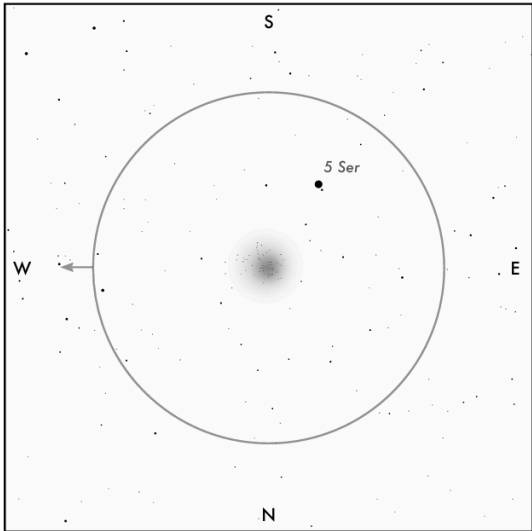
Star	Magnitude	Colour	Location
A (ζ^2)	4.96	Blue-white	Primary star
B (ζ^1)	5.91	Blue-white	6.3" NW from A

Both spectral class B dwarf stars, ζ^1 and ζ^2 (now understood to be a spectroscopic triple) are separated by ~ 900 AU and take more than 920 years to orbit their centre of mass. Locate it as the fourth-magnitude star west of Alkalurops in Boötes.

M5 (Globular Cluster) Easy

RA 15h 19m 54s | Dec 01° 59' 12" | Mag 5.69

Distant ~ 24,500 light years, M5 measures ~ 160 light years across and contains several hundred thousand stars (105 of which have been identified as variable). With low-mass stars that have begun expansion into red giants, it is one of the oldest known globular clusters, aged at 12 Gyr. M5 may be found by identifying the third magnitude star Unukalhai (Alpha Ser) and moving southeast. As one of the brightest globular clusters, it should be visible in the Dobsonian's finder scope. It appears with a slightly elliptical exterior, where hundreds of stars can be resolved at high powers, and a dense nucleus.



M16 - The Eagle Nebula (Emission Nebula) Medium

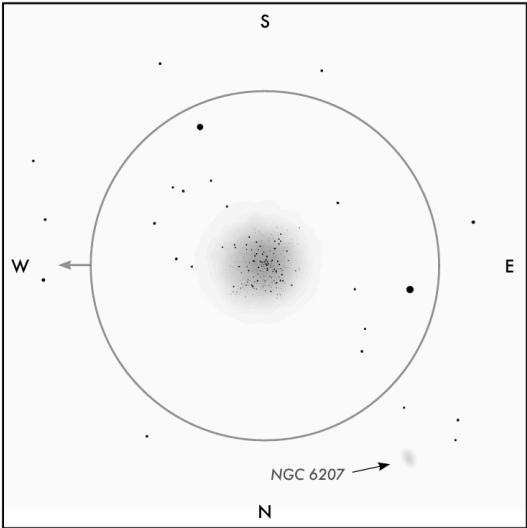
RA 18h 20m 22s | Dec -13° 49' 10" | Mag 6.40

Distant 5700 light years, M16 is perhaps best known for the 'Pillars of Creation' image produced using data from the Hubble Space Telescope. With an apparent diameter of 7', it is best observed visually at low power, where a few dozen individual stars of the cluster NGC 6611 can be resolved amongst the faint glow of nebulous concentrations. Good conditions and averted vision will help in observing the nebulous patches, but note that even the Herschels only catalogued the open cluster and not the nebula itself. (Photo credit: deepskywatch.com, Orion 8" eq. Newton.)

In the constellations Corona Borealis and Serpens, you might also like to observe:

- IC 4756 - Graff's Cluster (Open Cluster) Easy
- T CrB - Blaze Star (Variable Star) Medium
- Sigma CrB ((Visual) Double Star) Medium

Hercules



M13 - Hercules Great Globular Cluster Easy

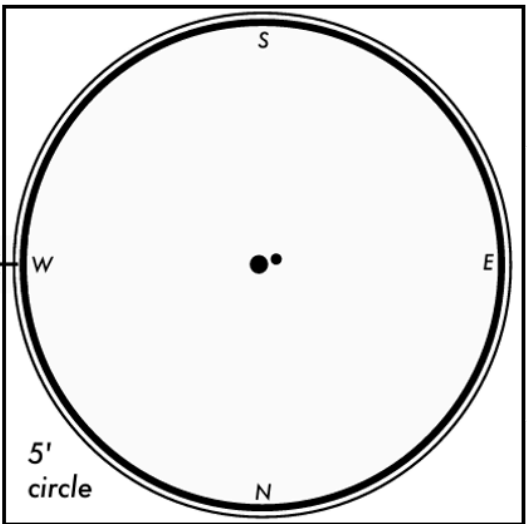
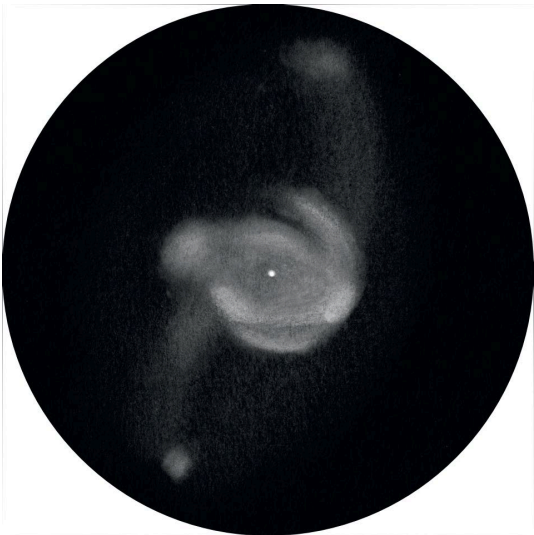
RA 16h 42m 39s | Dec 36° 24' 32" | Mag 5.80

M13 is arguably the best globular cluster to observe from northern latitudes, and can be found two thirds of the way along from Zeta Her to Eta Her, the two most western stars in the Keystone asterism. M13 contains more than 300,000 Population II stars and measures 145 light years in diameter. Its small yet bright core will be visible even on poor nights, and best viewing is at high power with a large aperture – try the Dobsonian with the Meade 9mm eyepiece. An incredible number of stars will be resolved towards the edges of the cluster, whilst the core will remain a dense, bright nucleus.

NGC 6210 - Turtle Nebula (Planetary Nebula) Hard

RA 16h 45m 37s | Dec 23° 44' 48" | Mag 8.80

NGC 6210 is a very small (0.3' by 0.3'), but fairly bright planetary nebula. Distant ~ 5400 light years, it measures only ~ 0.5 light years across and is a rather young planetary nebula, aged at ~ 2000 years. It contains a complex structure of shells and jets. Telescopes show a blue-green colour, due to the great surface brightness of the nebula, and so high magnifications are therefore ideal. To find it, pan your telescope 4° in a north-easterly direction from β Herculis. (Photo credit: *Uwe Glah*, 66 cm Newtonian telescope at 837 times magnification.)



Alpha Herculis - Rasalgethi (Double Star) Easy

RA 17h 15m 52s | Dec 14° 21' 33" | Mag 3.36, 5.40

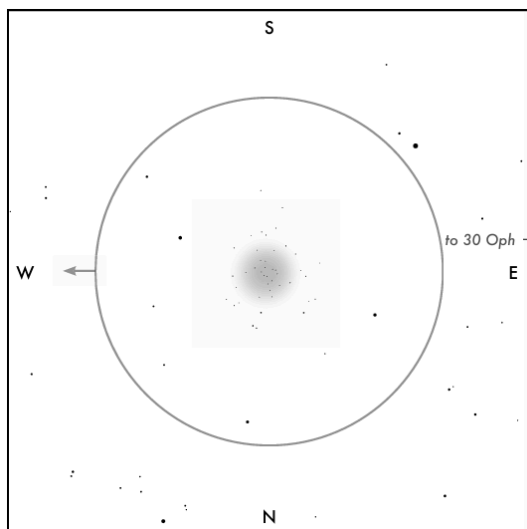
Star	Magnitude	Colour	Location
A	3.0 – 3.9	Red	Primary star
B	5.4	Yellow	4.8" ESE from A

The primary is an AGB star, a luminous red giant with a radius that pulsates irregularly over a three-month period leading to a variable magnitude. Distant more than 500 AU is the companion (itself a double), taking ~ 3600 years to orbit.

In the constellation Hercules, you might also like to observe:

- M92 (Globular Cluster) Easy
- Kappa Herculis - Marsic ((Optical) Double Star) Easy
- Delta Herculis - Serin ((Optical) Double Star) Medium

Ophiuchus (Globular Clusters)



M10 (Globular Cluster) Medium ▾

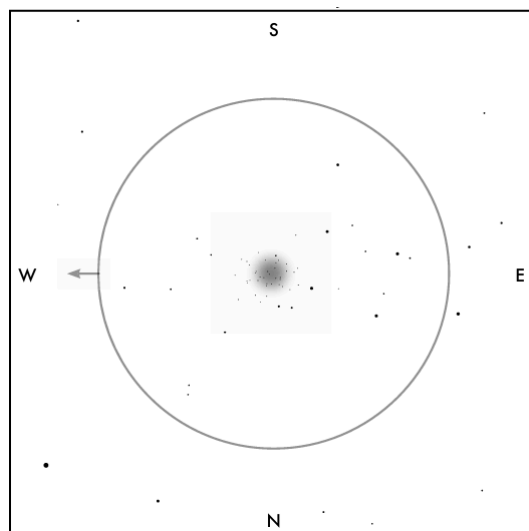
RA 16h 58m 34s | Dec -04° 08' 28" | Mag 6.59

M10 is located in the barren central region of Ophiuchus, at a distance of 14,300 light years. Its bright core measures 35 light years across and contains a significant concentration (~14%) of binary stars along with interaction-formed blue stragglers. Using the Dobsonian, locate the two third-magnitude stars which appear close to one another, Yed Prior and Yed Posterior, and move eastwards to find M10 situated to the west of the fifth-magnitude star 30 Oph. It will appear as a nebulous patch with individual stars being resolved at higher powers.

M12 (Globular Cluster) Medium ▾

RA 16h 48m 38s | Dec -01° 59' 44" | Mag 6.09

M12 is roughly 3° northwest of the globular cluster M10. Being rather loosely packed for a globular, M12 was once thought to have been a tightly concentrated open cluster. Distant 16,400 light years, in the Dobsonian, the granular, less concentrated core of M12 will appear 3' (14 light years) in diameter and is surrounded by a 10' (48 light years) halo of stars. Its brightest members are twelfth magnitude, and a 2006 study revealed a surprisingly small population of low-mass stars. It is thought that these stars were stripped away by the gravitational influence of the Milky Way.



M14 (Globular Cluster) Medium ▾

RA 17h 39m 01s | Dec -03° 15' 42" | Mag 7.59

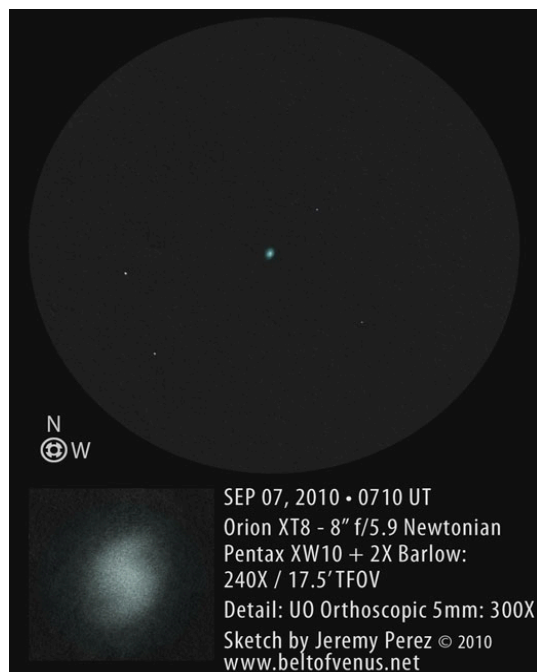
M14 is positioned east of the line connecting the stars Cebalrai and Sabik. Somewhat isolated from bright stars, M14 may be found by going 10° east from M10. Distant ~30,000 light years, some sources list an observable core diameter of ~3' (26 light years), and an elongated shape to the cluster (M14 appears like an elliptical galaxy, diffuse and faint, at first glance in a low power field). Whilst intrinsically much brighter than M10 and M12, its greater distance and position in a dusty region of the Milky Way, means it appears dimmer. (Photo credit: *deepskysketch.com*, from La Palma.)



In the constellation Ophiuchus, you might also like to observe:

- M9 (Globular Cluster) Hard ▾
- M107 (Globular Cluster) Hard ▾
- Barnard's Star (It is moving at a rate across the sky that is the fastest of any known star.)

Ophiuchus (cont.), Draco



NGC 6572 - Blue Racquetball (Planetary) Nebula Hard ▾

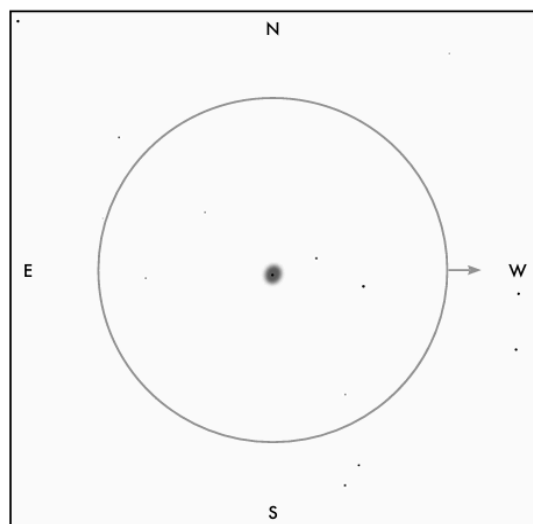
RA 18h 13m 24s | Dec 06° 51' 24" | Mag 8.00

NGC 6572 is a small planetary nebula with high surface brightness, the result of its youth and how the stellar material is still quite concentrated around the central star. Distant 6000 light years, it measures 0.3' by 0.2', and so will appear starlike at low magnifications. The intense blue-green colour of the nebula (due to the presence of O III) should help in its location, at which point a higher magnification should be used to reveal its shape. Currently, NGC 6572 boasts a surface brightness 100 times that of the Ring Nebula (M57) in Lyra, but as the gas shells expand, it will gradually dim and become more diffuse. To manually locate this nebula, begin at the fourth magnitude stars 71 Oph and 72 Oph and move a few degrees south. Located in a field of many fainter stars, NGC 6572 should be identifiable.

IC 4665 - Summer Beehive Cluster (Open Cluster) Easy ▾

RA 17h 47m 36s | Dec 05° 42' 21" | Mag 4.19

The Summer Beehive Cluster is an open cluster about 1° to the northeast of the star Beta Ophiuchi. It was discovered by Swiss astronomer Philippe Loys de Chéseaux in 1745, and is one of the brightest clusters not to be cataloged by Charles Messier or William Herschel. This is likely since IC 4665 is a very loose and coarse cluster. Like the Pleiades or the Hyades clusters, the best view is provided by binoculars or the finderscope of the Dobsonian telescope. It lies about 1100 light years away, contains 30 member stars and has an estimated age of 55 million years.



NGC 6543 - Cat's Eye Nebula (Planetary Nebula) Hard ▾

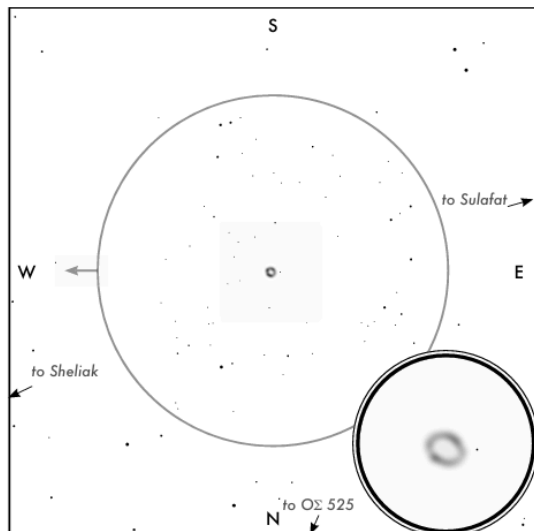
RA 17h 58m 33s | Dec 66° 37' 48" | Mag 8.10

NGC 6543 is a relatively young, yet greatly studied, planetary nebula, distant 3300 light years, with a highly complex structure of knots, jets and concentric gas shells. These shells that have been ejected by the central star over the past thousand years give resemblance to a cat's eye, though are difficult to discern visually. With a high Northern declination, refractor telescopes, such as the Northumberland and Thorowgood, are not suitable. Observed through a Dobsonian, say, NGC 6543 has a distinct blue-green colour and is best viewed using averted vision.

In the constellations Ophiuchus and Draco, you might also like to observe:

- Psi Draconis - Dziban ((Visual) Double Star) Easy ▾
- RY Draconis, UX Draconis (Carbon Stars) Easy ▾
- M102 - Spindle Galaxy Hard ▾

Lyra



M57 - Ring Nebula (Planetary Nebula) Easy ▾

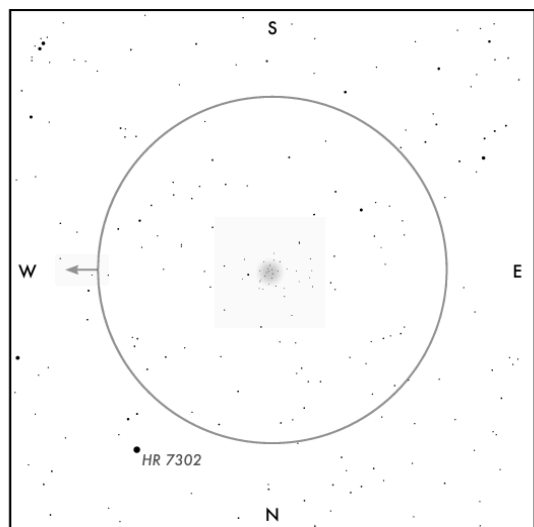
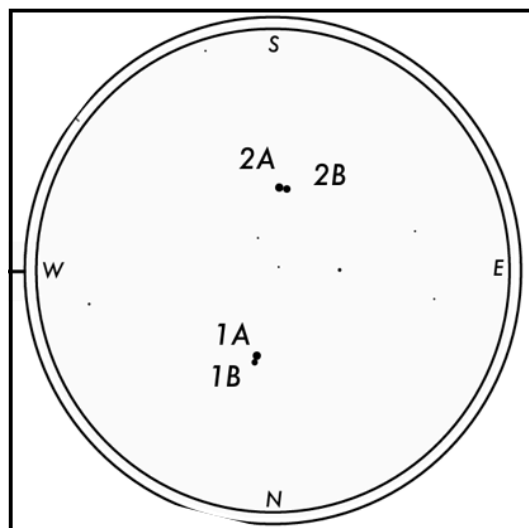
RA 18h 54m 35s | Dec 33° 03' 38" | Mag 8.80

As a planetary nebula, M57 (distant ~ 2,500 light years from Earth) is the result of the death of a low mass star forming a white dwarf. The UV radiation from the white dwarf ionises the gas, which then emits the light we observe. The distinct ring structure comes from the much higher density of oxygen and nitrogen in the outer layers compared with the helium core surrounding the white dwarf. (It is the concentrated levels of O III which dominates the surface brightness of the outer layer.) Expanding at 12 miles per second, it would have taken ~ 20,000 years for the nebula to reach its current size.

Epsilon Lyrae - Double Double (Multiple Star) Easy ▾

RA 18h 45m 13s | Dec 39° 41' 49" (Epsilon1 Lyrae)

With Vega (Alpha Lyrae) in the finder scope, observe a triangle of bright stars with Zeta Lyrae south of Vega and Epsilon Lyrae in the northwest. Separated by 209", ϵ Lyr can be easily resolved in the finder, but its two orbiting components, ϵ^1 Lyr and ϵ^2 Lyr, require higher magnifications. The components of both ϵ^1 and ϵ^2 Lyr are separated ~ 150 AU (2.2" and 2.4", respectively). The heavier components of ϵ^2 Lyr take ~ 600 years to orbit their centre of mass, whilst it takes more than 1000 years for ϵ^1 Lyr. The system is 160 light years distant, and consists of white stars of spectral type A.



M56 (Globular Cluster) Medium ▾

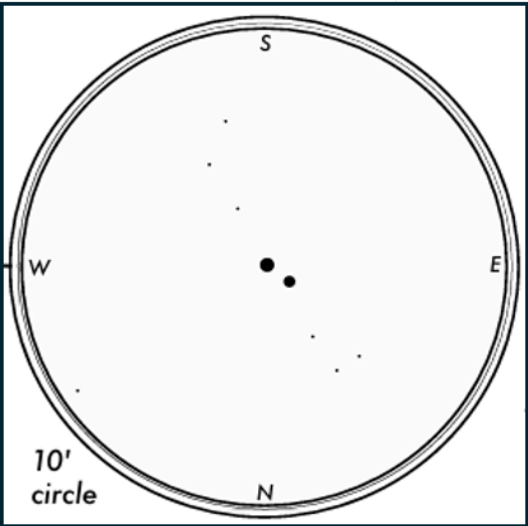
RA 19h 17m 38s | Dec 30° 13' 56" | Mag 8.39

Located roughly halfway between Albireo and Sulafat, M56 is one of the fainter Messier globulars and can be a difficult find. It lacks the characteristic bright core of most other globulars, but still contains more than 100,000 stars and measures ~ 85 light years across (8.8'). Sitting 32,900 light years away in a retrograde orbit through the Milky Way, it is thought that M56 was formed from a dwarf galaxy merger. The largest globular cluster, Omega Centauri (visible only in the southern hemisphere), would then form the surviving nucleus of this capture of a dwarf galaxy by the Milky Way.

In the constellation Lyra, you might also like to observe:

- Sheliak (Double Star) Easy ▾
- T Lyrae (Carbon Star) Medium ▾
- Struve 2470 and Struve 2474 ((Optical) Double Double Star System)) Medium ▾

Cygnus



Beta Cygni - Albireo (Double Star) Easy

RA 19h 31m 48s | Dec 28° 00' 55" | Mag 3.19, 4.67

Star	Magnitude	Colour	Location
A	3.19	Yellow-orange	Primary star
B	4.67	Blue-white	34" NE from A

Albireo is made of a giant, orange (spectral type K) star orbited by a hot, blue (spectral type B) star. Separated by at least 4000 AU, the companion takes ~ 100,000 years to orbit the primary - far too slow to observe any motion.

M29 - Cooling Tower Cluster (Open Cluster) Easy

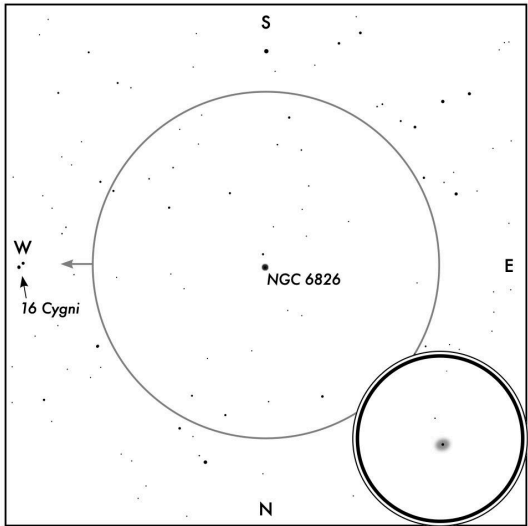
RA 20h 24m 56s | Dec 38° 35' 10" | Mag 6.59

Just south of Sadr (the central bright star in Cygnus) and within the nebulous, spiral arms of the Milky Way, is M29. Its five brightest members resemble the Plough asterism (a 'stubby dipper'), and are hot, blue stars of spectral class B0. The cluster is relatively young, ~ 10 Myr, and distant ~ 5000 light years.

M39 (Open Cluster) Easy

RA 21h 32m 46s | Dec 48° 33' 02" | Mag 4.59

Located ~ 9° NE of Deneb, M39 is a loosely-structured open cluster that is best observed at low powers (it spans a similar size to a full Moon). All of its members are main sequence stars. With the brightest at the point before evolution into the red giant phase, M39 has an estimated age of 270 Myr.



NGC 6826 - Blinking Planetary Nebula Hard

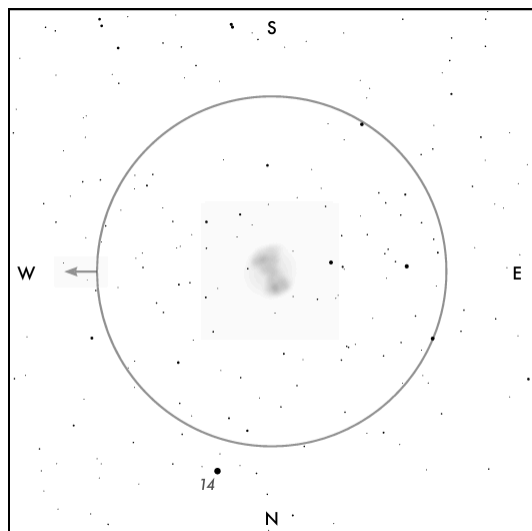
RA 19h 45m 31s | Dec 50° 35' 20" | Mag 8.89

The nebula shows clearly with averted vision and may disappear entirely if you look at it directly, since its light (equivalent to a ninth-magnitude star) is spread out over a small disc. Looking at it directly, you'll only be able to see the central star - a white dwarf left over from the collapse of a red giant. This "here again, gone again" feature of the nebula as the observer's eye wanders, gives it its name. It is located about 2000 light years away, and measures roughly 15,000 AU in diameter. It is a difficult find with the Dobsonian or Thorowgood, but can be reached with some star hopping.

In the constellation Cygnus, you might also like to observe:

- 16 Cygni (Double Star) Easy
- NGC 6960 - Veil Nebula (Supernova Remnant) Insane or Astrophotography (O III filter)
- NGC 6888 - Crescent Nebula (Emission Nebula) Insane or Astrophotography (O III filter)

Vulpecula, Sagitta, Delphinus



M27 - The Dumbbell (Planetary Nebula) Medium ▾

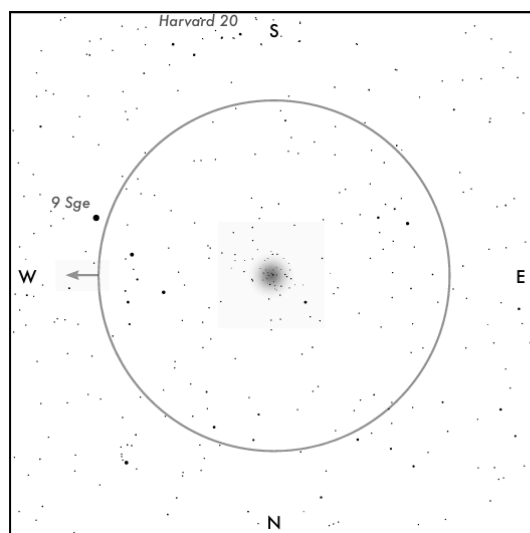
RA 20h 00m 45s | Dec 22° 47' 22" | Mag 7.09

M27 is formed from the collapse of a formerly Sun-like star to a white dwarf. The expanding gas shell, formed from the outer layers of the central star, ballooned into a pair of lobes above and below the plane of a ring of dense gas the star had previously ejected. M27 has a prolate spheroidal shape which, when viewed along the plane of the equator, means the two lobes are seen. Looking along the poles, M27 might bear a ring shape similar to M57 in Lyra. Extending two light years in diameter and expanding at a rate of 30 kilometers per second, the nebula has a kinematic age of 10,000 years.

M71 (Globular Cluster) Medium ▾

RA 19h 54m 58s | Dec 18° 50' 54" | Mag 8.39

For many decades, M71 was considered a densely packed open cluster, similar to M11 in Scutum. Its lack of RR Lyrae variable stars (used as standard candles) and high metallicity made the case for this, until the advent of modern photometry. The relative youth of M71, aged between 9 - 10 Gyr, also explains the abundance of 'metals'. (Most globulars formed from a giant molecular cloud in the very early universe before the first stars were able to go supernova, producing heavier elements.) Distant 13,000 light years, M71 can be easily located halfway between Gamma and Delta Sagittae.



Collinder 399 - Brocchi's Cluster (Asterism) Easy ▾

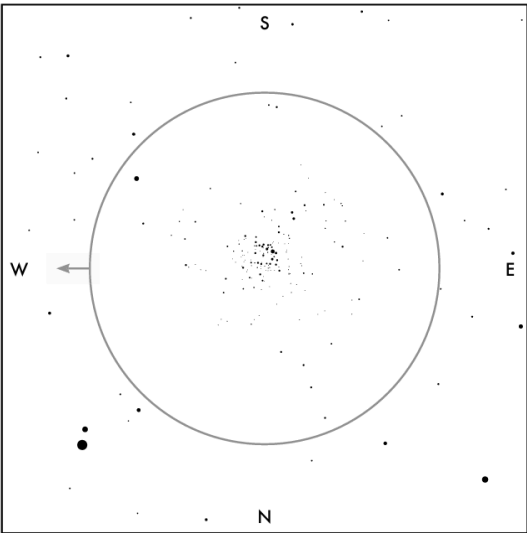
RA 19h 26m 34s | Dec 20° 14' 09" | Mag 3.59

At an apparent size of 89.0' (roughly three full Moons), Brocchi's Cluster is best viewed using binoculars or a finder scope and is a collection of fifth- to seventh-magnitude stars. Once thought to be an open cluster of gravitationally bound stars, it was not until 1997, when the Hipparcos satellite made measurements of stellar distances and proper motions, that the cluster was found to be a chance grouping of unrelated stars. Ranging from 200 to 2000 light years distant, the stars can be located a third of the way along from Altair to Vega. (Photo credit: Cloudy Nights 22/10/2007 WadeVC.)

In the constellations Vulpecula, Sagitta and Delphinus, you might also like to observe:

- Gamma Delphini ((Visual) Double Star) Easy ▾
- NGC 6823 (Open Cluster) Easy ▾ within NGC 6820 (Emission Nebula) Astrophotography ▾
- NGC 6934 (Globular Cluster) Hard ▾

Scutum, Aquila



M11 - The Wild Ducks (Open Cluster) Medium

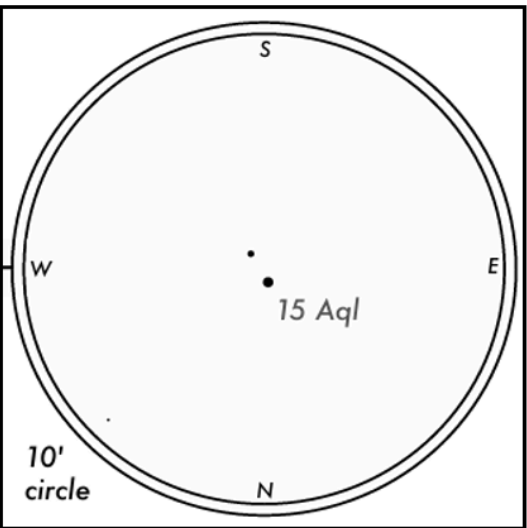
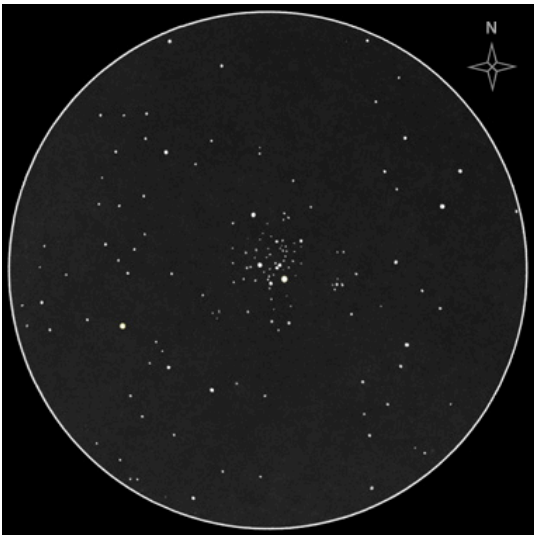
RA 18h 52m 31s | Dec -06° 14' 07" | Mag 5.80

M11 is best viewed at medium power (12 - 18 mm eyepiece) in a Dobsonian, as this will resolve the cluster into hundreds of individual stars against a background of grainy light. Only ~ 100 light years across with a core radius of 4.0 light years, the stars within M11 are, on average, less than one light year apart, making it one of the most compact open clusters. The brightest members are hot, young, blue and white stars of spectral class A and F. However, over a dozen stars have evolved to the red-giant stage, suggesting the cluster (6000 light years distant) is a few hundred millions years old.

M26 (Open Cluster) Medium

RA 18h 46m 46s | Dec -09° 21' 21" | Mag 8.00

One field of view in the finder scope southwest of M11, is the smaller and dimmer open cluster, M26. Distant 5000 light years and spanning 22 light years across, M26 is a collection of ~ 30 tenth-magnitude, and fainter, stars. Notably, there is a region of low star density near the nucleus of the cluster. This was thought to be the result of an obscuring cloud of interstellar matter between Earth and M26, but a paper, late last century, claimed that this was not possible. M26 is aged at 85.3 million years, and is best observed at medium power in a Dobsonian. (Photo credit: [deepskywatch.com](https://www.deepskywatch.com).)



15 Aquilae (Optical Double Star) Medium

RA 19h 06m 23s | Dec -03° 59' 29" | Mag 5.40, 6.98

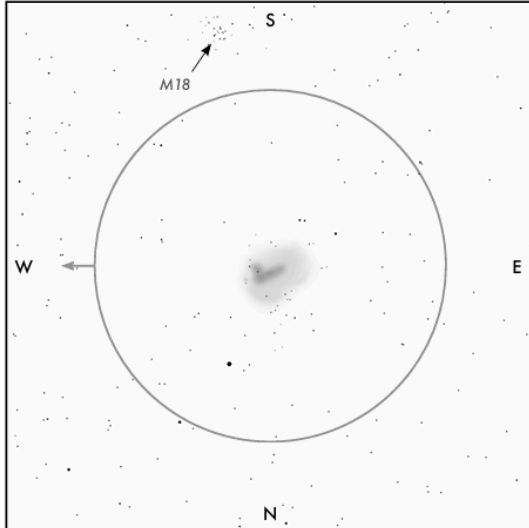
Star	Magnitude	Colour	Location
A	5.52	Yellow-orange	Primary star
B	6.98	Yellow-orange	40" SW from A

The primary is a giant, K-type star and is roughly four billion years old. With an orange hue similar to the primary star, the optical companion is much dimmer, but at a separation of 40", may be easily resolved in a small telescope.

In the constellations Scutum and Aquila, you might also like to observe:

- V Aquilae (Variable, Carbon Star) Medium
- NGC 6712 (Globular Cluster) Medium
- Delta Scuti (In 1.25 million years, it will be 9 light years from Earth, shining as bright as Sirius.)

Sagittarius



M17 - The Swan Nebula (Emission Nebula) Medium ▾

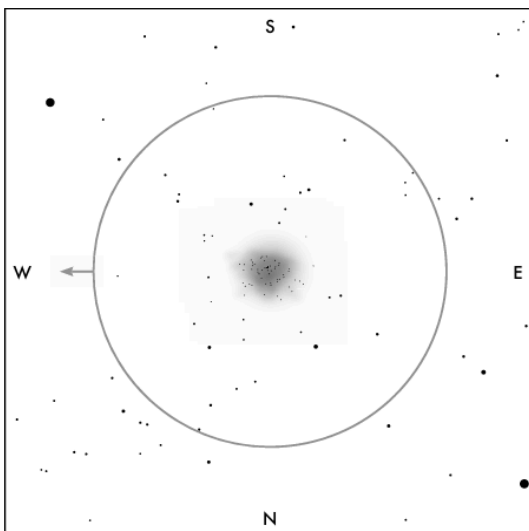
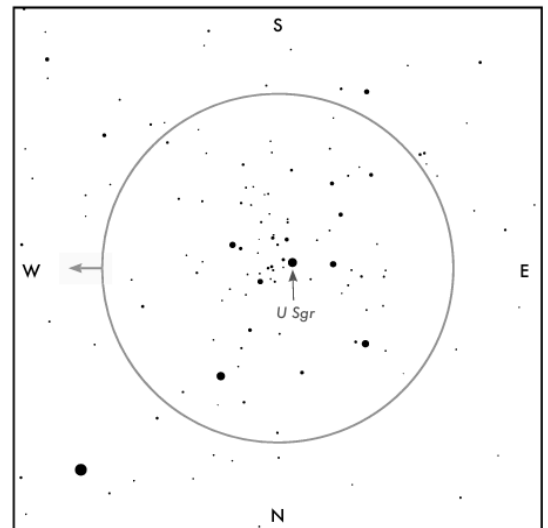
RA 18h 22m 37s | Dec -16° 09' 27" | Mag 6.00

M17 is an H II region: a molecular cloud consisting of ~ 90% partially ionised hydrogen gas, with helium forming the rest. It is distant 5000 - 6000 light years from Earth and extends ~ 40 light years across. The embedded open cluster NGC 6618 causes the gases of the nebula to shine due to radiation emitted from its hot, young stars. With a total mass of ~ 800 solar masses, M17 is one of the brightest and most massive star-forming regions in the Milky Way, with a similar local geometry to M42, the Orion Nebula. The 'check mark' bar of light within the nebula is an easily discernible feature of M17.

M25 (Open Cluster) Hard ▾

RA 18h 33m 21s | Dec -19° 05' 52" | Mag 4.59

M25 is 2000 light years distant from Earth, and is one of the more impressive open clusters in Sagittarius. In a Dobsonian at medium power (as in the figure on the right), it is a widely spaced grouping of about 60 stars, with many being quite bright. Notably, M25 contains the Cepheid variable star U Sgr, which varies between magnitudes 6.3 and 7.0 over a period of 6.75 days. Such stars are important for accurate distance measurements. Also within M25 are two stars of spectral type G. These well-evolved, red giants suggest M25 is not a very young cluster, aged at 67.6 millions years old.



M22 (Globular Cluster) Hard ▾

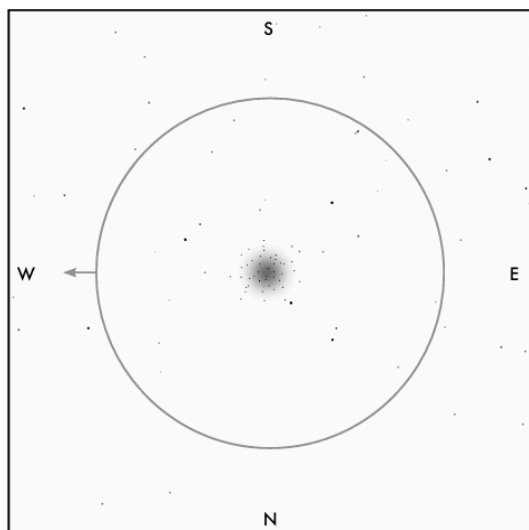
RA 18h 38m 02s | Dec -23° 52' 51" | Mag 5.19

M22 is one of the brightest globular clusters in the sky with a distinctly elliptical shape. Lying just below the ecliptic, however, its position in the sky from Cambridge means it can be challenging to observe. Distant 10,600 light years from Earth, M22 is one of the nearer globular clusters, and has a diameter close to 100 light years. As shown on the left, in a Dobsonian at medium power, the brightest members of M22 are easily resolved in the asymmetric ball of light. M22 is one of only four globular clusters to contain a planetary nebula, which was discovered using IRAS and is only 6000 years old.

In the constellation Sagittarius, you might also like to observe:

- M20 - Trifid Nebula (Emission Nebula) [Insane or Astrophotography ▾](#)
- M23 (Open Cluster) [Hard ▾](#)
- M24 - Sagittarius Star Cloud [Hard ▾](#)

Aquarius



M2 (Globular Cluster) Medium ▾

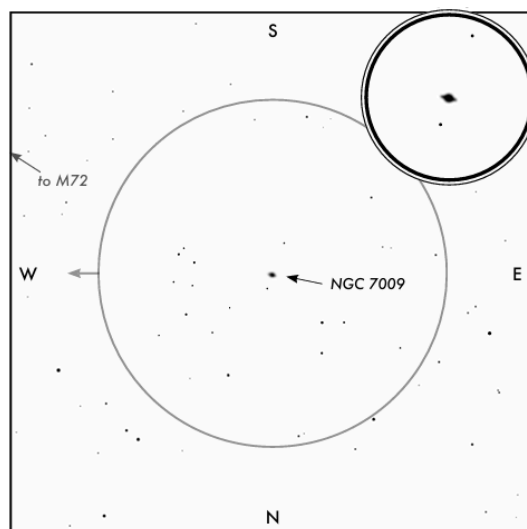
RA 21h 34m 50s | Dec -00° 42' 15" | Mag 6.59

With a diameter of 175 light years, M2 is one of the largest globular clusters known. At medium power in a Dobsonian, it is distinctly elliptical in the northwest-southeast direction with a modestly intense core and many individual stars that can be resolved across the disk. Using averted vision, several dark lanes (due to obscuring clouds of interstellar dust) may be observed, most prominently, in the NE of the cluster. Distant 37,500 light years, M2 also has an estimated age of 13.2 billion years (Botev et al., 2025) and is part of the Gaia Sausage (the remains of a dwarf galaxy merger).

NGC 7009 - Saturn Nebula (Planetary Nebula) Hard ▾

RA 21h 05m 38s | Dec -11° 15' 34" | Mag 7.80

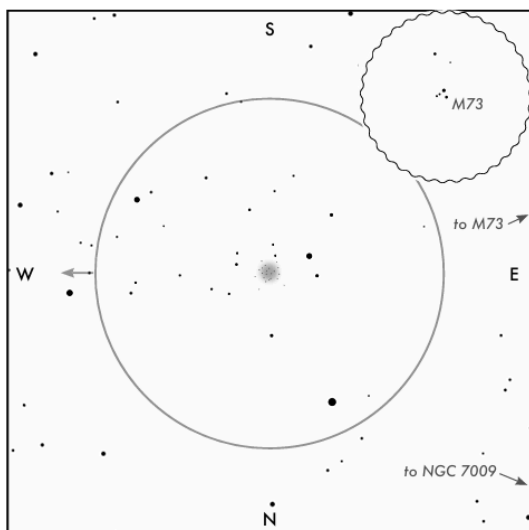
At low power, NGC 7009 is almost starlike, but with a distinct bluish-green hue, due to radiation emitted from doubly ionised oxygen created by the intense UV radiation emitted by the central white dwarf. The inset in the righthand image shows a high power (5') field of view of the Saturn Nebula, where short, faint equatorial ansae ('handles') can be seen to extend from the major axis, creating an impression of the planetary rings observed around Saturn. With no reference stars in the neighbourhood, the distance to NGC 7009 is not well known. Best estimates are of a few thousand years.



M72 (Globular Cluster) Hard ▾

RA 20h 54m 56s | Dec -12° 26' 07" | Mag 9.19

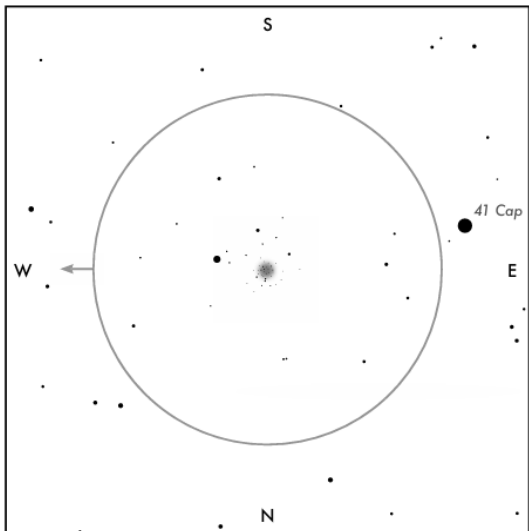
M72, one of the younger globular clusters, aged at 9.5 billion years, is loosely concentrated, with no sharp core, appearing as a pale, nebulous patch of light. Its fifteenth-magnitude stars are difficult to resolve, even with the large apertures of the Northumberland and Dobsonian. Distant 54,500 light years away, in a sparse region of the sky, M72 is a suitably difficult object to try star hopping to. Along the way, the collection of four, unrelated stars that constitute M73, may be located roughly 1.5° to the east of M72. M73, similarly to Brocchi's Cluster, was, at first, thought to be an open cluster.



In the constellation Aquarius, you might also like to observe:

- KV Aquarii (Variable, Double Star) Medium ▾
- M73 (Asterism) Hard ▾
- NGC 7293 - Helix Nebula (Planetary Nebula) Insane or Astrophotography ▾

Capricornus, Pegasus, Cepheus

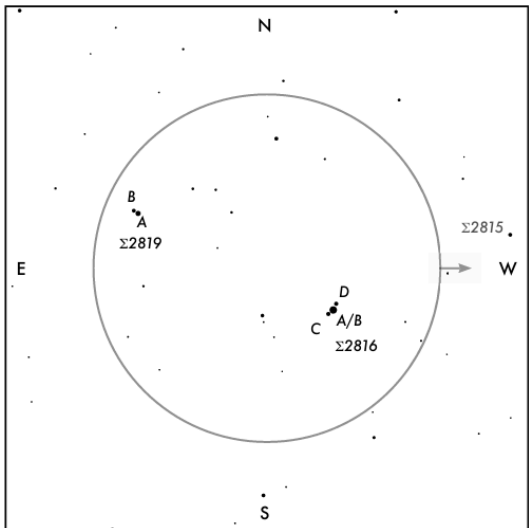
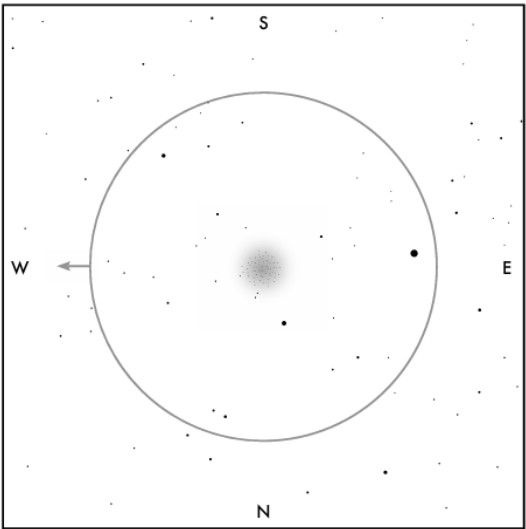


M30 (Globular Cluster) Medium
RA 21h 41m 53s | Dec -23° 03' 26" | Mag 6.90

M30, positioned just below the ecliptic, is longest observable during the first half of August when it is positioned opposite the Sun. Rising to a maximum of ~ 15° above the horizon in Cambridge, it may still be a challenge to observe. In a low power field, M30 can be located as a dim disk to the west of the fifth-magnitude star, 41 Cap. The cluster extends to a diameter of 12', where individual stars may be resolved at high magnifications, and collapses to a core 1' wide with an extremely dense stellar population. With M30 distant 27,000 light years, its core measures less than one light year across.

M15 (Globular Cluster) Easy
RA 21h 31m 16s | Dec 12° 17' 06" | Mag 6.30

As with M30, M15 has undergone a contraction, referred to as core collapse, resulting in a highly dense and bright nucleus that then thins rapidly from the centre. Higher magnifications will resolve individual stars, with dark lanes and patches prominent on the northeast edge of the cluster's inner shell. Distant 36,000 light years from Earth, M15 is also noted for being a source of X-rays (thought to be from supernova remnants, rather than a central black hole), and it contains the first planetary nebula, Pease 1, to be discovered within a globular cluster.



Struve 2816 and Struve 2819 (Multiple Stars) Medium
RA 21h 39m 47s | Dec 57° 36' 35" (Struve 2816)

Star	Magnitude	Colour	Location
2816 A	5.73	Yellow	Primary
C	7.48	White	12" ESE from A
D	7.53	White	20" NW from A
2819 A	7.44	Yellow-white	Primary
B	8.64	Yellow-white	13" NE from A

In the constellations Capricornus, Pegasus and Cepheus, you might also like to observe:

- Mu Cephei - Garnet Star Easy
- Delta Cephei (*The Cepheid Variable Star*) Easy
- NGC 7317, 7318 A and B, 7319, 7320 - Stephan's Quintet (Galaxies) Astrophotography

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